

THE THREE ERAS OF AUTONOMOUS COMPUTING

From Structured Industrial Determinism to Cloud AI and Embodied Physical Agents

ERA 1: CLASSICAL AUTOMATION

1970s – 2010s

Rigid, Deterministic Execution

- Structured factory workcells & caged robots
- Hard-coded ladder logic & PID controllers
- Zero open-world generalization capability
- Safe only through physical human exclusion

ERA 2: CLOUD & DIGITAL AI

2012 – 2024

Disembodied Statistical Scaling

- Large Language Models & Generative Diffusion
- Pure software sandbox · Infinite undo & idempotent APIs
- Unbounded tail latencies (> 500 ms acceptable)
- Zero awareness of momentum, mass, or heating

ERA 3: PHYSICAL AI

2024 – Present

Embodied Real-Time Agency

- Multi-rate dual-brain architecture (MPU $\leftrightarrow$ MCU)
- Foundation world models with real-time CBF filters
- Hard latency freshness deadlines ($d_{\text{stop}} \leq d_{\text{gap}}$)
- Irreversible physical consequence & zero software undo